Phạm Hải Đăng

Backend Engineer

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SUMMARY

Backend Engineer with ~1 year building real-time multiplayer game backends. Independently designed and operated a 14-service production MMORPG (NestJS + Go) running 24/7. Strong in concurrency control, transaction reliability, and system scalability.

WORK EXPERIENCE

Mobifos JSC - Gahu Studio 8/2025 - 5/2026

Fullstack Developer (Part-time)

- Developed 3 NestJS microservices for a casual match-3 puzzle game handling **~2000 CCU**, using REST and gRPC
- Built RabbitMQ async pipelines with retry (3x) and exponential backoff, recovering **~30% of transient failures** in email and data sync
- Optimized data access using MySQL indexing and Redis caching, **reducing average query latency by ~70%**
- Hardened services with JWT/RBAC, rate limiting, Docker, Winston logging, and Swagger

Gamota - Gabros Studio 11/2025 - 2/2026

Backend Developer (Internship)

- Designed and implemented a real-time multiplayer backend using WebSocket, load-tested to **1000+ CCU** with low-latency updates
- Replaced real-time DB writes (20/s) with Redis-backed state and RabbitMQ async batch persistence, **cutting DB write load by ~95%**
- Designed idempotent and transaction-safe workflows, handling distributed transactions, race conditions, and consistency issues
- Ensured atomic in-game item purchase using **Redis + Lua scripting** to prevent race conditions under high concurrency

PROJECT

Sacred Dragon Warriors – Production MMORPG 2/2025 - Now

Fullstack Engineer

Overview

- Designed and operated a **14-service** microservices backend (NestJS + Go) for a live MMORPG running **24/7** over 14 months, with 1000+ commits across 17 repos (client+server) and a CI/CD pipeline shipping **478 deployments** at **95% success rate**.

Architecture

- Polyglot microservices (Go + NestJS) using REST, gRPC, WebSocket; event-driven via RabbitMQ, Redis Pub/Sub, NATS and BullMQ, **100+ APIs across 8 databases** (PostgreSQL/MySQL for transactional, MongoDB for logging, Redis for caching/locking).

Technical Highlights

- Used Saga + Outbox pattern for distributed transactions — **0% partial-failure incidents** across all transactional flows
- Optimized performance via Dirty Flag + async batch writes **(~90% DB write reduction)** and Fire-and-Forget (~10ms latency saved), maintaining **700–1,000 RPS** throughput
- Integrated PayOS and VietQR with idempotent webhook handling, achieving **100% transaction accuracy** across ~100 test cases
- Migrated the real-time game loop from NestJS/Socket.IO to Go/raw WebSocket + Protobuf + NATS, **reducing payload size by ~60%**
- Integrated RAG chatbot for in-game player support with Redis-cached FAQ responses (**~40% cache hit rate**, **~50% LLM cost reduction**) and per-user rate limiting
- Built observability stack using Prometheus, Grafana, PM2 monitoring, circuit breakers, and Discord/Telegram alerting; tracked service health, latency, and resource metrics, detecting 25 production incidents.
- Deployed across a 3-node Ubuntu VPS cluster with load-balanced stateless services and horizontal scaling.
- Automated daily backup and restore (cron 04:00) for MySQL, PostgreSQL, MongoDB, and Redis with versioned snapshots, **7-day retention**, across local, GitHub, and Google Drive.
- Soak-tested at **1,000 RPS** with **p99 <500ms, 98.4% success rate**; stress-tested up to 3,000 VUs (k6) with system saturation at ~1,200 RPS

Project Links: https://ngocrongdark.com (live demo) | https://github.com/DANG-PH/dragonboy-api-gateway (server)

TECHNICAL SKILLS

Programming Languages	TypeScript, JavaScript, Python, Java, Go
System Design	Saga, Outbox, Idempotency, Concurrency Control, Eventual Consistency
Backend & Architecture	Microservices, REST, gRPC, WebSocket, Distributed Systems, Event-driven
Databases & Messaging	MySQL, PostgreSQL, MongoDB, Redis, RabbitMQ, NATS, BullMQ
Security	JWT, RBAC, OAuth2, 2FA, Rate Limiting, CORS, Validation, XSS Sanitize
DevOps & Infrastructure	Docker, Nginx, Cloudflare, CI/CD, VPS, PM2

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT) 2024 - 2028

Information Technology

Second-year student at PTIT with GPA: **3.3 / 4.0**

HONORS & AWARDS

Top 16, RMIT AI & Security Hackathon	2025
Freshman Year Academic Scholarship	2025
Second Prize, Provincial High School Mathematics Olympiad	2024